Analyzing a Downtrending Department and Proposing a Solution

Lukas Ishihara
Business Analyst, The Collective
lukas.ishihara@gmail.com

Business Question

Is the department transaction total declining month-over-month and what is the solution to turn the department around?

Department's Transaction Decline

- Trend:
  - Confirmed a downtrend with a loss of 1.6M in transaction sales month-over-month.
- Structural Decline:
  - Analysis of historical data yielded a R^2 of .18 which indicates that time is <u>not a large factor</u> and is instead a structural decline.

PostgreSQL

Analysis performed in PostgreSQL

Identifying the Parameters

User Segmentation

The top 10% of platform derived income earners are 7x as active in this department as the other 90% combined

Profit Margin

With the current product mix, the business makes 8.3%. However, the business has a 3% floor in their lowest margin game and cannot be raised to prevent user alienation and thus decrease sales.

Declined Proposal

A cashback reward system is not likely to meaningfully produce an increase since the largest daily user contributed 345x in sales of that of the median. Most of the current sales are dominated by a small number of users.

Analysis performed in SQLite

Business Recommendation:

- Offer a set of VIP roles and leaderboards for users that actively engage in the department's sales for thresholds of >0, >100k, >500k.
    - Users consistently vie for the social status of being at the top of the business's other leaderboards.
- Do not instate a reward program.
    - Rationale: The profit margins are already low and cannot be raised to prevent user alienation and thus a decrease in sales.

Appendix: Slide 1 of 3

Methodology & Simulation Logic

Analytical Process

- Trend Analysis
  - Performed Linear Regression and found R^2
- Defined Parameters of the Proposal
  - Identified the target market
  - Analyzed Profit Margin and Per-Product Value
- Hypothesized a Proposal for a Cashback Reward System and Simulated the Outcome
  - Used a rolling 30 days to model the profits of the optimal and systemic worst-case scenarios as the range
  - Found the required sales multiplier that the business required for the range
- Performed a User Behavior Analysis
  - Used historic data to determine the percent of users that would be likely incentivized
  - Rejected the hypothesized proposal
- Ultimately, Recommended a New Low-Cost Solution

Why I Rejected a Model Supported by Math

- Simulation: A cashback program required only a 6% sales increase to break even.
- Reality: Users likely to be incentivized was 1-2 amongst thousands of users.
- The Conflict:
  - The target was too high for 99% of the population
  - The target was too low to properly incentivize the 1%
- Result: The business cannot rely on incentivizing 1-2 people while frustrating everyone else